

2024考研数学真题题解

数学二

一、选择题1~10, 每题5分

1、(2024-2)函数 $f(x) = |x|^{\frac{1}{(1-x)(x-2)}}$ 第一类间断点个数().

A. 3; B. 2; C. 1; D. 0.

解: $f(x) = e^{\frac{1}{(1-x)(x-2)} \ln|x|}$ 的间断点有0,1,2.

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} e^{\frac{1}{(1-x)(x-2)} \ln|x|} = \lim_{x \rightarrow 0} e^{\frac{-1}{2} \ln|x|} = e^{+\infty} = +\infty \quad x=0 \text{ 无穷间断点}$$

$$\lim_{x \rightarrow 1} f(x) = \lim_{x \rightarrow 1} e^{\frac{1}{(1-x)(x-2)} \ln x} = \lim_{x \rightarrow 1} e^{\frac{\frac{1}{x}}{(2-x)+(1-x)}} = e \quad x=1 \text{ 可去间断点}$$

$$\lim_{x \rightarrow 2} f(x) = \lim_{x \rightarrow 2} e^{\frac{1}{(1-x)(x-2)} \ln x} = \begin{cases} \lim_{x \rightarrow 2^+} e^{\frac{1}{(1-x)(x-2)} \ln x} = e^{-\infty} = 0 & x=2 \text{ 无穷间断} \\ \lim_{x \rightarrow 2^-} e^{\frac{1}{(1-x)(x-2)} \ln x} = e^{+\infty} = \infty \end{cases}$$

只有一个第一类间断点

选C

2、(2024-2)已知 $\begin{cases} x=1+t^3 \\ y=e^{t^2} \end{cases}$, 则 $\lim_{x \rightarrow +\infty} x \left[f\left(2+\frac{2}{x}\right) - f(2) \right] = (\quad)$.

A. $2e$; B. $\frac{4}{3}e$; C. $\frac{2}{3}e$; D. $\frac{1}{3}e$

解: 导数定义与参量方程求导结合

$$\lim_{x \rightarrow +\infty} x \left[f\left(2+\frac{2}{x}\right) - f(2) \right] = \lim_{x \rightarrow +\infty} \frac{\left[f\left(2+\frac{2}{x}\right) - f(2) \right]}{\frac{2}{x}} \cdot 2 = 2f'_+(2)$$

$x=2$ 时, $t=1$

$$\frac{dy}{dx} \Big|_{t=1} = \frac{2te^{t^2}}{3t^2} \Big|_{t=1} = \frac{2e}{3}$$

$$\text{因此 } \lim_{x \rightarrow +\infty} x \left[f\left(2+\frac{2}{x}\right) - f(2) \right] = \frac{4}{3}e$$

选B

3、(2024-1/2)已知函数 $f(x) = \int_0^x e^{\cos t} dt$, $g(x) = \int_0^{\sin x} e^{t^2} dt$, 则().

- A. $f(x)$ 是奇函数, $g(x)$ 是偶函数 B. $f(x)$ 是偶函数, $g(x)$ 是奇函数
C. $f(x)$ 与 $g(x)$ 都是奇函数 D. $f(x)$ 与 $g(x)$ 都是偶函数

解：方法一：连续的奇函数的原函数是偶函数，

而连续的偶函数的原函数中只有 $\int_0^x f(t) dt$ 是奇函数。

求导： $f'(x) = \left(\int_0^x e^{\cos t} dt \right)' = e^{\cos x}$ $g'(x) = \left(\int_0^{\sin x} e^{t^2} dt \right)' = e^{\sin^2 x} \cos x$

显然 $f'(x), g'(x)$ 均为偶函数，因此 $f(x), g(x)$ 均为奇函数. **C正确**

方法二：利用定义 $f(-x) = \int_0^{-x} e^{\cos t} dt \stackrel{v=-t}{=} - \int_0^x e^{\cos(-v)} dv = - \int_0^x e^{\cos v} dv = -f(x)$

类似的 $g(-x) = \int_0^{\sin(-x)} e^{t^2} dt \stackrel{v=-t}{=} - \int_0^{\sin x} e^{(-v)^2} dv = - \int_0^{\sin x} e^{v^2} dv = -g(x).$

4、(2024-2)已知数列 $\{a_n\}$ ($a_n \neq 0$), 若 $\{a_n\}$ 发散, 则().

A. $\left\{a_n + \frac{1}{a_n}\right\}$ 发散; B. $\left\{a_n - \frac{1}{a_n}\right\}$ 发散; C. $\left\{e^{a_n} + \frac{1}{e^{a_n}}\right\}$ 发散; D. $\left\{e^{a_n} - \frac{1}{e^{a_n}}\right\}$ 发散.

解: 举反例 首先举例 $a_n = (-1)^n$ $\{a_n\} : -1, 1, -1, 1 \dots -1, 1$

因为 $a_n - \frac{1}{a_n} \equiv 0$ 所以 $\left\{a_n - \frac{1}{a_n}\right\}$ 收敛 B 不对

因为 $e^{a_n} + \frac{1}{e^{a_n}} = e^{a_n} + e^{-a_n} \equiv e + e^{-1}$ 所以 $\left\{e^{a_n} + \frac{1}{e^{a_n}}\right\}$ 收敛 C 不对

举例 $a_n = 2^{(-1)^n}$ $\frac{1}{2}, 2, \frac{1}{2}, 2 \dots \frac{1}{2}, 2$ 则 $a_n + \frac{1}{a_n} \equiv \frac{5}{2}$ 即 $\left\{a_n + \frac{1}{a_n}\right\}$ 收敛 A 不对

排除法 D 正确

5、(2024-2)已知函数 $f(x, y) = \begin{cases} (x^2 + y^2) \sin \frac{1}{xy} & xy \neq 0 \\ 0 & xy = 0 \end{cases}$,则在点(0,0)处().

A. $\frac{\partial f(x, y)}{\partial x}$ 连续, $f(x, y)$ 可微;

B. $\frac{\partial f(x, y)}{\partial x}$ 连续, $f(x, y)$ 不可微;

C. $\frac{\partial f(x, y)}{\partial x}$ 不连续, $f(x, y)$ 可微;

D. $\frac{\partial f(x, y)}{\partial x}$ 不连续, $f(x, y)$ 不可微.

解: 考察多元函数偏导存在、可微、偏导连续关系

x, y 地位对称

$$\lim_{x \rightarrow 0} \frac{f(x, 0) - f(0, 0)}{x} = \lim_{x \rightarrow 0} \frac{0 - 0}{x} = 0 = f'_x(0, 0)$$

类似的 $f'_y(0, 0) = 0$, 偏导存在

$$xy \neq 0 \text{ 时, } \frac{\partial f(x, y)}{\partial x} = 2x \sin \frac{1}{xy} + (x^2 + y^2) \cos \frac{1}{xy} \cdot \frac{1}{y} \cdot \frac{-1}{x^2}$$

特殊的

$$\lim_{y=x \rightarrow 0} \left(2x \sin \frac{1}{xy} + (x^2 + y^2) \cos \frac{1}{xy} \cdot \frac{1}{y} \cdot \frac{-1}{x^2} \right) \text{ 不存在} \quad \lim_{(x, y) \rightarrow (0, 0)} \frac{\partial f(x, y)}{\partial x} \neq f'_x(0, 0)$$

由 $\lim_{(x,y) \rightarrow (0,0)} \frac{\partial f(x,y)}{\partial x} \neq f'_x(0,0)$ $\frac{\partial f(x,y)}{\partial x}$ 不连续

偏导存在：是否可微等价于 $\lim_{(x,y) \rightarrow (0,0)} \frac{f(x,y) - f(0,0) - f'_x(0,0)x - f'_y(0,0)y}{\sqrt{x^2 + y^2}} = 0$

而 $\lim_{(x,y) \rightarrow (0,0)} \frac{f(x,y) - f(0,0) - f'_x(0,0)x - f'_y(0,0)y}{\sqrt{x^2 + y^2}} = \lim_{(x,y) \rightarrow (0,0)} \frac{(x^2 + y^2) \sin \frac{1}{xy}}{\sqrt{x^2 + y^2}}$

$= \lim_{(x,y) \rightarrow (0,0)} \sqrt{x^2 + y^2} \sin \frac{1}{xy} = 0$ $f(x,y)$ 在 $(0,0)$ 处可微

C正确

6、(2024-2/3) 设 $f(x, y)$ 是连续函数, 则 $\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} dx \int_{\sin x}^1 f(x, y) dy = (\quad)$.

A. $\int_{\frac{1}{2}}^1 dy \int_{\frac{\pi}{6}}^{\arcsin y} f(x, y) dx;$

B. $\int_{\frac{1}{2}}^1 dy \int_{\arcsin y}^{\frac{\pi}{2}} f(x, y) dx;$

C. $\int_0^{\frac{1}{2}} dy \int_{\frac{\pi}{6}}^{\arcsin y} f(x, y) dx;$

D. $\int_0^{\frac{1}{2}} dy \int_{\arcsin y}^{\frac{\pi}{2}} f(x, y) dx.$

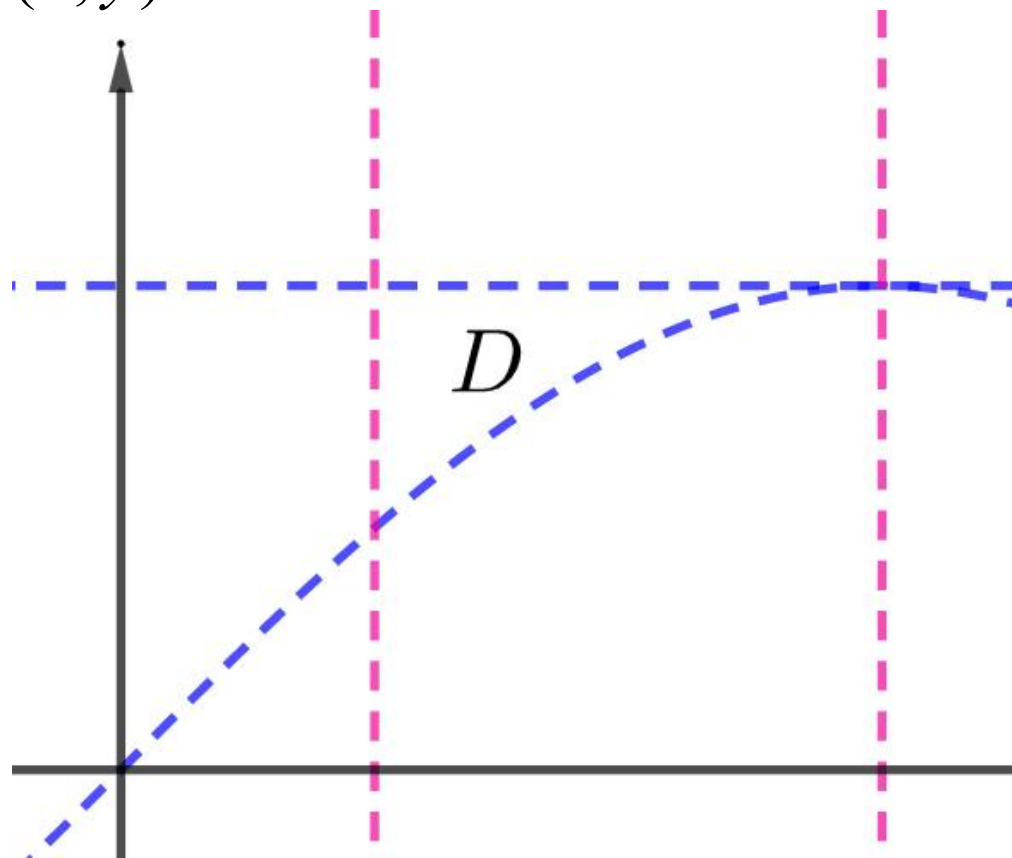
解:

交换积分次序

$$D = \begin{cases} \frac{\pi}{6} \leq x \leq \frac{\pi}{2} \\ \sin x \leq y \leq 1 \end{cases}$$

$$D = \begin{cases} \frac{1}{2} \leq y \leq 1 \\ \frac{\pi}{6} \leq x \leq \arcsin y \end{cases}$$

A正确



7、(2024-2) 设非负函数 $f(x)$ 在 $[0, +\infty)$ 上连续, 给定以下三个命题:

(1) 若 $\int_0^{+\infty} f^2(x) dx$ 收敛, 则 $\int_0^{+\infty} f(x) dx$ 收敛;

(2) 若存在 $p > 1$ 使 $\lim_{x \rightarrow +\infty} x^p f(x)$ 存在, 则 $\int_0^{+\infty} f(x) dx$ 收敛;

(3) 若 $\int_0^{+\infty} f(x) dx$ 收敛, 则存在 $p > 1$, 使极限 $\lim_{x \rightarrow +\infty} x^p f(x)$ 存在.

A. 0; B. 1; C. 2; D. 3.

解: 举反例 命题(1) $f(x) = \frac{1}{x+1}$ $\int_0^{+\infty} f(x) dx$ 发散, $\int_0^{+\infty} f^2(x) dx$ 收敛 命题(1)错

命题(3) $f(x) = \frac{1}{(x+2)\ln^2(x+2)}$ $\int_0^{+\infty} f(x) dx$ 收敛 但 $\lim_{x \rightarrow +\infty} x^p \frac{1}{(x+2)\ln^2(x+2)} = \infty$ $p > 1$

命题(2)是结论 $\lim_{x \rightarrow +\infty} x^p f(x) = a$ $f(x) = \frac{a}{(x+1)^p} + o\left(\frac{1}{(x+1)^p}\right)$ $\int_0^{+\infty} f(x) dx$ 收敛

B正确

8、(2024-2/3) 设 A 为 3 阶矩阵, $P = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$, 若 $P^T A P^2 = \begin{pmatrix} a+2c & 0 & c \\ 0 & b & 0 \\ 2c & 0 & c \end{pmatrix}$, 则 $A = (\quad)$.

A. $\begin{pmatrix} c & 0 & 0 \\ 0 & a & 0 \\ 0 & 0 & b \end{pmatrix}$; B. $\begin{pmatrix} b & 0 & 0 \\ 0 & c & 0 \\ 0 & 0 & a \end{pmatrix}$; C. $\begin{pmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{pmatrix}$; D. $\begin{pmatrix} c & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & a \end{pmatrix}$.

解: $P^T = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ $(P^T)^{-1} = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ $P^2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{pmatrix}$ $(P^2)^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{pmatrix}$

$$A = (P^T)^{-1} \begin{pmatrix} a+2c & 0 & c \\ 0 & b & 0 \\ 2c & 0 & c \end{pmatrix} (P^2)^{-1} = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} a+2c & 0 & c \\ 0 & b & 0 \\ 2c & 0 & c \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{pmatrix} = \begin{pmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 2c & 0 & c \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{pmatrix}$$

C正确

9、(2024-2) 设 A 为 4 阶矩阵, A^* 为 A 伴随矩阵, 若 $A(A - A^*) = O$, 且 $A \neq A^*$, 则 $r(A)$ 取值为().

A. 0 或 1; B. 1 或 3; C. 2 或 3; D. 1 或 2.

解: $\because A(A - A^*) = O$

$$\therefore r(A) + r(A - A^*) \leq 4 \quad ①$$

又 $\because A \neq A^*$

$$\therefore r(A - A^*) \geq 1 \quad ②$$

由 ①② 得 $1 \leq r(A) \leq 3 \quad ③$

由 ③ 可知 A 不满秩 $\Leftrightarrow |A| = 0$

可得 $A^2 = AA^* = |A|E = 0$

即 $r(A) + r(A) \leq 4 \quad ④$

由 ③④ 可得 $1 \leq r(A) \leq 2$

D 正确

10、(2024-2) 设 A, B 均为 2 阶矩阵, 且 $AB = BA$, 则 “ A 有两个不相等的特征值” 是 “ B 可对角化” 的 () 条件.

A. 充要; B. 充分不必要; C. 必要不充分; D. 既不充分也不必要.

解: 设 α 为 A 的对应特征值 λ 的特征向量, 即 $A\alpha = \lambda\alpha$

左乘 B 矩阵, $BA\alpha = B\lambda\alpha$ 由 $AB = BA$ $AB\alpha = \lambda B\alpha$

① 若 $B\alpha \neq 0$ $B\alpha$ 为 A 的属于特征值 λ 的特征向量

属于同一特征值的特征向量成比例 $B\alpha = k\alpha$ α 为 B 的特征向量

② 若 $B\alpha = 0$ $B\alpha = 0 \cdot \alpha$ α 为 B 的特征向量

A 有两个不相等特征值 $\Rightarrow A$ 有两个不相关特征向量

B 也有两个不相关特征向量 B 可对角化 “前” $\Rightarrow$ “后”

反过来, 举例 $A = E = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, B = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$ B 正确

事实上, B 可对角化连 B 具有两个不相等特征值都推不出来

二、填空题11~16, 每题5分

11、(2024-2)曲线 $y^2 = x$ 在点(0,0)处曲率圆方程为_____.

解: 对 $y^2 = x$ 两边关于 x 求导

$$2yy' = 1 \Rightarrow y' = \frac{1}{2y}$$

$$\text{关于} x \text{再求导} 2(y')^2 + 2yy'' = 0 \quad y'' = \frac{2(y')^2}{2y} = \frac{-1}{4y^3}$$

$$\text{曲率} k = \frac{|y''|}{\left(1 + (y')^2\right)^{\frac{3}{2}}} = \frac{\frac{1}{4y^3}}{\left(1 + \frac{1}{4y^2}\right)^{\frac{3}{2}}} = \frac{2}{(1 + 4y^2)^{\frac{3}{2}}}$$

$$y = 0, k = 2 \quad \text{曲率圆半径} R = \frac{1}{2}$$

$$\text{曲率圆方程} \left(x - \frac{1}{2}\right)^2 + y^2 = \frac{1}{4}$$

12、(2024-2/3)函数 $f(x, y) = 2x^3 - 9x^2 - 6y^4 + 12x + 24y$ 的极值点为_____.

解：求驻点 $\begin{cases} f'_x(x, y) = 6x^2 - 18x + 12 = 0 \\ f'_y(x, y) = -24y^3 + 24 = 0 \end{cases}$ 驻点(1,1)、(2,1)

$$f''_{xx}(x, y) = 12x - 18, f''_{xy}(x, y) = 0, f''_{yy}(x, y) = -72y^2$$

对于驻点(1,1)处 $\therefore \begin{vmatrix} -6 & 0 \\ 0 & -72 \end{vmatrix} = 432 > 0$, 且 $f'_{xx}(1,1) < 0$ $\therefore (1,1)$ 是极大值点

对于驻点(2,1)处 $\therefore \begin{vmatrix} 6 & 0 \\ 0 & -72 \end{vmatrix} = -432 < 0$ $\therefore (2,1)$ 不是极值点

极值点(1,1)

13、(2024-1/2)微分方程 $y' = \frac{1}{(x+y)^2}$ 满足条件 $y(1) = 0$ 的解为 _____.

解： 做变量替换 令 $x + y = u$ 则 $1 + \frac{dy}{dx} = \frac{du}{dx}$

原方程可化为 $\frac{du}{dx} - 1 = \frac{1}{u^2}$ 分离变量 $\frac{u^2 du}{u^2 + 1} = dx$

积分可得 $u - \arctan u = x + c$

即 $y - \arctan(x + y) = c$ 带入 $y(1) = 0$, 得 $c = -\frac{\pi}{4}$

$$y = \arctan(x + y) - \frac{\pi}{4}$$

$$\text{或 } x = \tan\left(y + \frac{\pi}{4}\right) - y$$

14、(2024-2)已知 $f(x) = x^2(e^x + 1)$, 则 $f^{(5)}(1) = \underline{\hspace{2cm}}$.

解: 莱布尼茨乘法高阶导公式

$$\left[x^2(e^x + 1) \right]^{(5)} = C_5^0(e^x + 1)^{(5)} \cdot x^2 + C_5^1(e^x + 1)^{(4)} \cdot 2x + C_5^2(e^x + 1)^{(3)} \cdot 2$$

$$= x^2 e^x + 10x e^x + 20e^x$$

$$\text{带入 } x=1 \quad f^{(5)}(1) = e + 10e + 20e = 31e$$

15、(2024-2)某物体以速度 $v(t) = t + k \sin \pi t$ 做直线运动.若从 $t = 0$ 到 $t = 3$ 时间内平均速度是 $\frac{5}{2}$, 则 $k =$ _____.

解: 积分中值定理, $\overline{f(x)} = \frac{1}{b-a} \int_a^b f(x) dx$

$$\frac{5}{2} = \overline{v(t)} = \frac{1}{3} \int_0^3 v(t) dt = \frac{1}{3} \int_0^3 (t + k \sin \pi t) dt$$

$$= \frac{1}{3} \left(\frac{t^2}{2} - \frac{k}{\pi} \cos \pi t \right) \Big|_0^3 = \frac{1}{3} \left(\frac{9}{2} + \frac{2k}{\pi} \right) \quad \text{化简} \frac{15}{2} = \frac{9}{2} + \frac{2k}{\pi}$$

$$k = \frac{3\pi}{2}$$

16、(2024-1/2) 设向量 $\alpha_1 = \begin{pmatrix} a \\ 1 \\ -1 \\ 1 \end{pmatrix}, \alpha_2 = \begin{pmatrix} 1 \\ 1 \\ b \\ a \end{pmatrix}, \alpha_3 = \begin{pmatrix} 1 \\ a \\ -1 \\ 1 \end{pmatrix}$, 若 $\alpha_1, \alpha_2, \alpha_3$ 线性相关,

任意两个无关, 则 $a \cdot b = \underline{\hspace{2cm}}$. (数学一选择6)

解: 由 $\alpha_1, \alpha_2, \alpha_3$ 线性相关且任意两个无关 $r(\alpha_1, \alpha_2, \alpha_3) = 2$

矩阵 $(\alpha_1, \alpha_2, \alpha_3)$ 中任意三阶子式为零

$$\text{即 } \begin{vmatrix} a & 1 & 1 \\ 1 & 1 & a \\ 1 & a & 1 \end{vmatrix} = (a+2) \begin{vmatrix} 1 & 1 & 1 \\ 1 & 1 & a \\ 1 & a & 1 \end{vmatrix} = -(a+2)(a-1)^2 = 0$$

若 $a = 1$

α_1, α_3 相关

$a = -2$

$$\text{又 } \begin{vmatrix} -2 & 1 & 1 \\ 1 & 1 & -2 \\ -1 & b & -1 \end{vmatrix} = \begin{vmatrix} 0 & 3 & -3 \\ 1 & 1 & -2 \\ 0 & b+1 & -3 \end{vmatrix} = - \begin{vmatrix} 3 & -3 \\ b+1 & -3 \end{vmatrix} = 0$$

$b = 2$

$a \cdot b = -4$

17、(2024-2/3) 设平面区域 D 位于第一象限, 由曲线 $xy = \frac{1}{3}$, $xy = 3$, 直线 $y = \frac{1}{3}x$, $y = 3x$

围成, 计算 $\iint_D (1+x-y) dx dy$. $= \frac{8}{3} \ln 3$

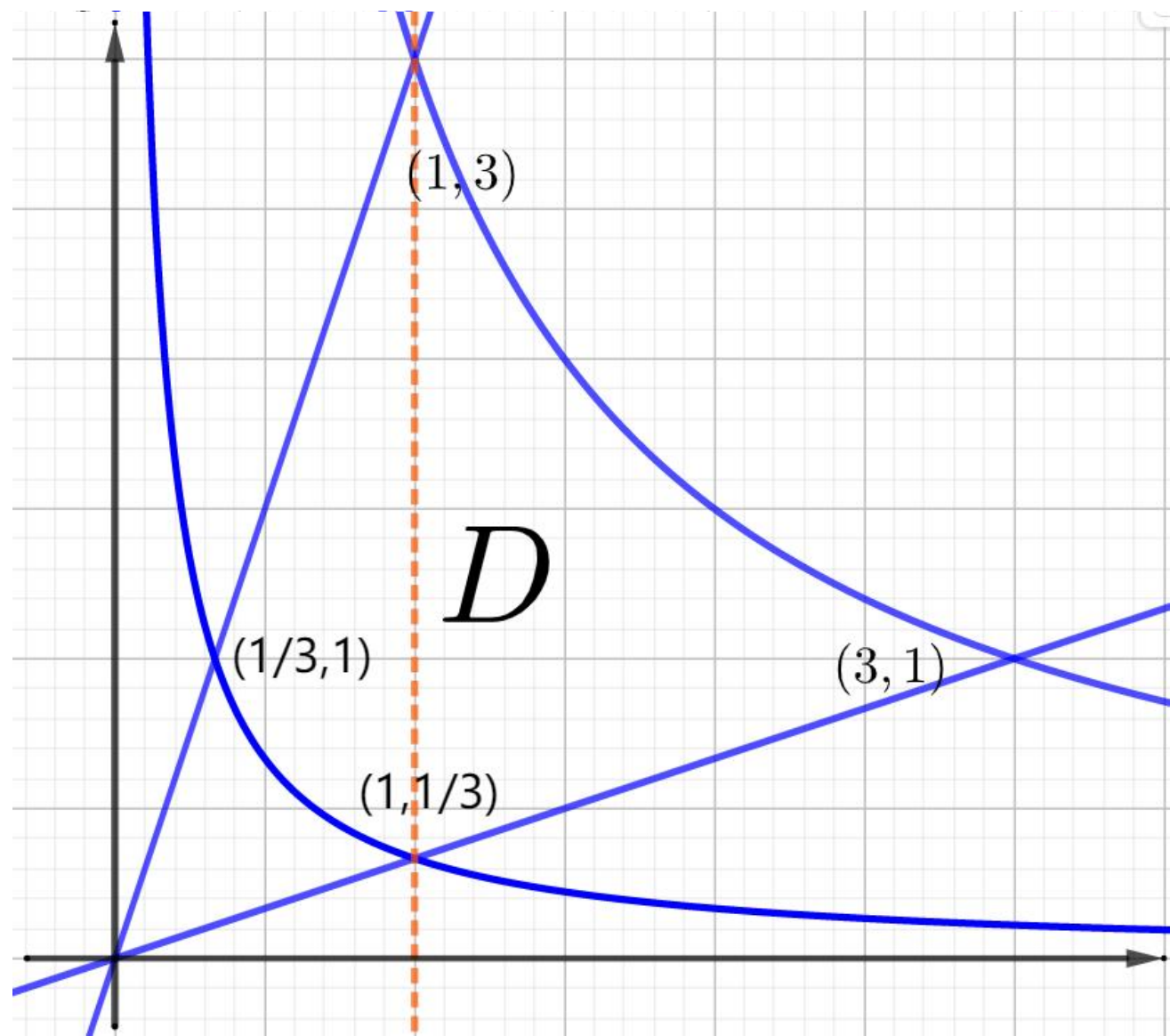
解: 区域 D 关于 $y = x$ 对称

由轮换对称性 $\iint_D x dx dy = \iint_D y dx dy$

因此 $\iint_D (1+x-y) dx dy = \iint_D 1 dx dy = S_D$

根据 D 的特点, 方法一, 直角下计算

$$\begin{aligned} S_D &= \int_{\frac{1}{3}}^1 \left(3x - \frac{1}{3x}\right) dx + \int_1^3 \left(\frac{3}{x} - \frac{x}{3}\right) dx \\ &= \left(\frac{3x^2}{2} - \frac{\ln|x|}{3}\right) \Big|_{\frac{1}{3}}^1 + \left(3\ln|x| - \frac{x^2}{6}\right) \Big|_1^3 \\ &= 3/2 - (1/6 + \ln 3/3) + 3\ln 3 - 3/2 - (-1/6) \end{aligned}$$

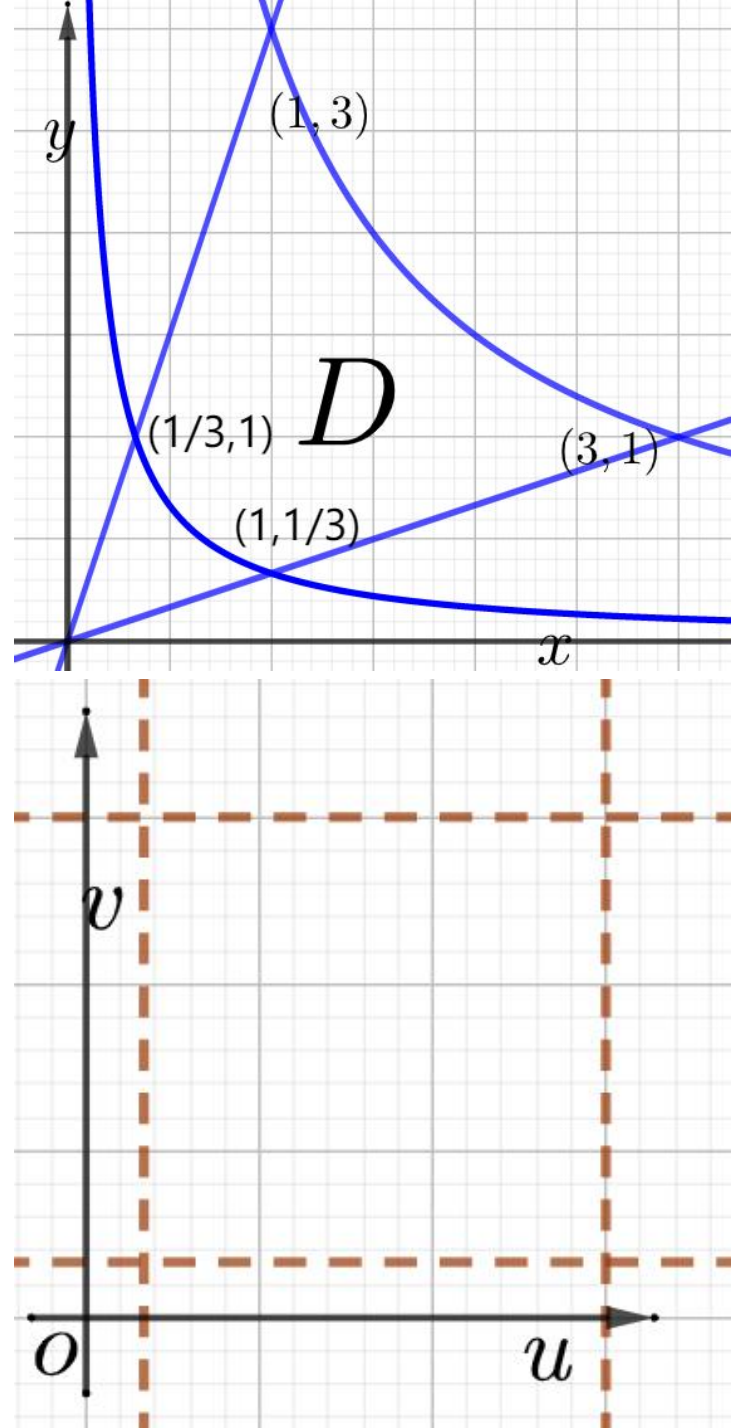


轮换对称 $\iint_D (1+x-y) dx dy = \iint_D 1 dx dy = S_D$

方法二进行坐标变换 令 $\begin{cases} xy = u \\ \frac{y}{x} = v \end{cases} \begin{cases} \frac{1}{3} \leq u \leq 3 \\ \frac{1}{3} \leq v \leq 3 \end{cases}$ 则 $\begin{cases} x = \sqrt{\frac{u}{v}} \\ y = \sqrt{uv} \end{cases}$

$$\frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} \frac{1}{\sqrt{v}} \cdot \frac{1}{2\sqrt{u}} & \sqrt{u} \frac{1}{v} \cdot \frac{-1}{2\sqrt{v}} \\ \sqrt{v} \frac{1}{2\sqrt{u}} & \sqrt{u} \frac{1}{2\sqrt{v}} \end{vmatrix} = \frac{1}{2v}$$

$$\begin{aligned} S_D &= \iint_{D_{xy}} 1 dx dy = \iint_{D_{uv}} \left| \frac{\partial(x,y)}{\partial(u,v)} \right| du dv = \int_{\frac{1}{3}}^3 du \int_{\frac{1}{3}}^3 \frac{1}{2v} dv \\ &= \frac{8}{3} \left(\frac{1}{2} \ln v \Big|_{\frac{1}{3}}^3 \right) = \frac{4}{3} \left(\ln 3 - \ln \frac{1}{3} \right) = \frac{8}{3} \ln 3 \end{aligned}$$



18、(2024-2) 设 $y = y(x)$ 满足 $x^2 y'' + xy' - 9y = 0$, $y|_{x=1} = 2$, $y'|_{x=1} = 6$. (1) 利用变换 $x = e^t$

化简方程, 并求 $y(x)$ 表达式; (2) 求 $\int_1^2 y(x) \sqrt{4-x^2} dx$.

解: (1) 欧拉方程 $x = e^t, t = \ln x$ 微分算子 D 表示: $Dy = \frac{dy}{dt}$

$$xy' = Dy, x^2 y'' = D(D-1)y$$

原方程可化为: $D(D-1)y + Dy - 9y = 0$ 即 $\frac{d^2 y}{dt^2} - 9y = 0$

特征方程即 $r^2 - 9 = 0$

$$y(t) = c_1 e^{3t} + c_2 e^{-3t} = c_1 x^3 + \frac{c_2}{x^3} \quad \text{带入定解条件: } \begin{cases} c_1 + c_2 = 2 \\ 3c_1 - 3c_2 = 6 \end{cases} \quad y(x) = 2x^3$$

$$\begin{aligned} (2) \int_1^2 y(x) \sqrt{4-x^2} dx &= \int_1^2 2x^3 \sqrt{4-x^2} dx = \int_1^2 x^2 \sqrt{4-x^2} dx^2 = \int_1^4 t \sqrt{4-t} dt \\ &= \int_{\sqrt{4-t}}^0 (4-v^2)v \cdot (-2v) dv = 2 \int_0^{\sqrt{3}} (4v^2 - v^4) dv = 2 \left(\frac{4v^3}{3} - \frac{v^5}{5} \right) \Big|_0^{\sqrt{3}} = 2 \left(4\sqrt{3} - \frac{9\sqrt{3}}{5} \right) \\ &= \frac{22}{5} \sqrt{3} \end{aligned}$$

19、(2024-2) 设 $t > 0$, 平面有界区域 D 由 $y = \sqrt{x}e^{-x}$ 与两直线 $x = t, x = 2t$ 及 x 轴所围成, D 绕 x 轴旋转一周所成旋转体体积为 $V(t)$, 求 $V(t)$ 最大值.

$$\begin{aligned}\text{解: } V(t) &= \int_t^{2t} \pi y^2 dx = \pi \int_t^{2t} x e^{-2x} dx = \frac{-\pi}{2} \int_t^{2t} x d e^{-2x} \\ &= \frac{-\pi}{2} x e^{-2x} \Big|_t^{2t} + \frac{\pi}{2} \int_t^{2t} e^{-2x} dx = \frac{-\pi}{4} (4t e^{-4t} + e^{-4t} - 2t e^{-2t} - e^{-2t}) \quad \textcircled{1}\end{aligned}$$

$$V'(t) = \pi (4t e^{-4t} - t e^{-2t}) = \pi t e^{-2t} (4e^{-2t} - 1) \quad V'(t) = 0 \Rightarrow t = \ln 2$$

$$V''(t) = \pi (4e^{-4t} - 16t e^{-4t} - e^{-2t} + 2t e^{-2t}) \quad e^{-4 \ln 2} = \frac{1}{16}, e^{-2 \ln 2} = \frac{1}{4}$$

$$V''(\ln 2) = \pi \left(\frac{1}{4} - \ln 2 - \frac{1}{4} + \frac{1}{2} \ln 2 \right) = -\frac{\ln 2}{2} \pi < 0$$

$$\text{由①式, 最大值 } V(\ln 2) = \frac{-\pi}{4} \left(\frac{\ln 2}{4} + \frac{1}{16} - \frac{\ln 2}{2} - \frac{1}{4} \right) = \frac{\pi}{16} \left(\ln 2 + \frac{3}{4} \right)$$

20、(2024-2)已知函数 $f(u, v)$ 具有二阶连续偏导, 且函数 $g(x, y) = f(2x + y, 3x - y)$

满足 $\frac{\partial^2 g}{\partial x^2} + \frac{\partial^2 g}{\partial x \partial y} - 6 \frac{\partial^2 g}{\partial y^2} = 1$. (1) 求 $\frac{\partial^2 f}{\partial u \partial v}$;

$$f(u, v) = \frac{uv}{25} - (u+1)e^{-u} + \frac{v^2}{50}$$

(2) 若 $\frac{\partial f(u, 0)}{\partial u} = ue^{-u}$, $f(0, v) = \frac{v^2}{50} - 1$, 求 $f(u, v)$ 表达式.

解: (1) $\frac{\partial g}{\partial x} = f'_u \cdot 2 + f'_v \cdot 3$ $\frac{\partial g}{\partial y} = f'_u \cdot 1 - f'_v \cdot 1$ (2) 由 $f''_{uv} = \frac{1}{25}$ 则 $f'_u = \frac{v}{25} + c(u)$

$$\frac{\partial^2 g}{\partial x^2} = (f''_{uu} \cdot 2 + f''_{uv} \cdot 3) \cdot 2 + (f''_{vu} \cdot 2 + f''_{vv} \cdot 3) \cdot 3$$

带入 $f'_u(u, 0) = c(u) = ue^{-u}$

$$= 4f''_{uu} + 12f''_{uv} + 9f''_{vv}$$

即 $f'_u(u, v) = \frac{v}{25} + ue^{-u}$

$$\frac{\partial^2 g}{\partial x \partial x} = 2f''_{uu} + f''_{uv} - 3f''_{vv} \quad \frac{\partial^2 g}{\partial y^2} = f''_{uu} - 2f''_{uv} + f''_{vv}$$

则 $f(u, v) = \frac{uv}{25} - (u+1)e^{-u} + d(v)$

带入 $\frac{\partial^2 g}{\partial x^2} + \frac{\partial^2 g}{\partial x \partial y} - 6 \frac{\partial^2 g}{\partial y^2} = 1$ 得 $25f''_{uv} = 1$ $f''_{uv} = \frac{1}{25}$

带入 $f(0, v) = -1 + d(v) = \frac{v^2}{50} - 1$

显然 $d(v) = \frac{v^2}{50}$

21、(2024-1/2/3) 设函数 $f(x)$ 具有二阶导数, 且 $f'(0) = f'(1)$, $|f''(x)| \leq 1$, 求证

$$(1) \text{ 当 } x \in (0,1) \text{ 时, } |f(x) - f(0)(1-x) - f(1)x| \leq \frac{x(1-x)}{2};$$

$$(2) \left| \int_0^1 f(x) dx - \frac{f(0) + f(1)}{2} \right| \leq \frac{1}{12}.$$

证明: (1) 将 $f(x)$ 在 0,1 分别进行泰勒展开

$$\begin{cases} f(x) = f(0) + f'(0)x + \frac{f''(\xi_1)}{2!}x^2 \end{cases} \quad (1)$$

$$\begin{cases} f(x) = f(1) + f'(1)(x-1) + \frac{f''(\xi_1)}{2!}(x-1)^2 \end{cases} \quad (2)$$

$$(1) \times (1-x) + (2) \times x \quad f(x) = f(0)(1-x) + f(1)x + \frac{f''(\xi_1)}{2!}(1-x)x^2 + \frac{f''(\xi_1)}{2!}x(x-1)^2$$

$$\begin{aligned} |f(x) - f(0)(1-x) - f(1)x| &= \left| \frac{f''(\xi_1)}{2!}(1-x)x^2 + \frac{f''(\xi_1)}{2!}x(x-1)^2 \right| \\ &\leq \frac{1}{2} |(1-x)x^2 + x(x-1)^2| \leq \frac{(1-x)x}{2} \end{aligned}$$

(2)证明:

$$\text{由(1)可知: } -\frac{x(1-x)}{2} \leq f(x) - f(0)(1-x) - f(1)x \leq \frac{x(1-x)}{2} \quad \textcircled{3}$$

将③在 $[0,1]$ 上积分

$$-\int_0^1 \frac{x(1-x)}{2} dx \leq \int_0^1 [f(x) - f(0)(1-x) - f(1)x] dx \leq \int_0^1 \frac{x(1-x)}{2} dx \quad \textcircled{4}$$

$$\text{计算} \int_0^1 [f(x) - f(0)(1-x) - f(1)x] dx = \int_0^1 \left[f(x) - \frac{f(0) + f(1)}{2} \right] dx$$

$$\text{计算右边} = \frac{1}{2} \int_0^1 (x - x^2) dx = \frac{1}{12}$$

$$\text{则④化为 } \left| \int_0^1 f(x) dx - \frac{f(0) + f(1)}{2} \right| \leq \frac{1}{12}.$$

22.(2024-2) 设矩阵 $A = \begin{pmatrix} 0 & 1 & a \\ 1 & 0 & 1 \end{pmatrix}$, $B = \begin{pmatrix} 1 & 1 \\ b & 2 \end{pmatrix}$, $f(x_1, x_2, x_3) = x^T B A x$,

已知方程组 $Ax = 0$ 的解是 $B^T x = 0$ 的解, 但两个方程不同解.

(1) 求 a, b 值; (2) 求正交变换 $x = Qy$, 将 $f(x_1, x_2, x_3)$ 化为标准型.

解: (1) 由题意知 $Ax = 0$ 与 $\begin{pmatrix} A \\ B^T \end{pmatrix} x = 0$ 同解 即 $r\begin{pmatrix} A \\ B^T \end{pmatrix} = r(A) = 2$

$$\begin{pmatrix} A \\ B^T \end{pmatrix} = \begin{pmatrix} 0 & 1 & a \\ 1 & 0 & 1 \\ 1 & 1 & b \\ 1 & 1 & 2 \end{pmatrix} \rightarrow \begin{pmatrix} 0 & 1 & a \\ 1 & 0 & 1 \\ 0 & 1 & b-1 \\ 0 & 1 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} 0 & 1 & a \\ 1 & 0 & 1 \\ 0 & 0 & b-a-1 \\ 0 & 0 & 1-a \end{pmatrix}$$

因此 $\begin{cases} b-a-1=0 \\ 1-a=0 \end{cases}$

即 $\begin{cases} a=1 \\ b=2 \end{cases}$

$$(2) \text{ 令 } C = BA = \begin{pmatrix} 1 & 1 \\ 1 & 1 \\ 2 & 2 \end{pmatrix} \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 2 \\ 1 & 1 & 2 \\ 2 & 2 & 4 \end{pmatrix} \quad \text{即 } f(x_1, x_2, x_3) = x^T Cx$$

$$|\lambda E - C| = \begin{vmatrix} \lambda - 1 & -1 & -2 \\ -1 & \lambda - 1 & -2 \\ -2 & -2 & \lambda - 4 \end{vmatrix} = \begin{vmatrix} \lambda - 1 & -1 & -2 \\ -1 & \lambda - 1 & -2 \\ 0 & -2\lambda & \lambda \end{vmatrix} = \begin{vmatrix} \lambda - 1 & -5 & -2 \\ -1 & \lambda - 5 & -2 \\ 0 & 0 & \lambda \end{vmatrix} = \lambda^2(\lambda - 6)$$

特征值 $\lambda_1 = \lambda_2 = 0, \lambda_3 = 6$ 分别求对应的特征向量

$$(0 \cdot E - C)x = 0 \quad \begin{pmatrix} -1 & -1 & -2 \\ -1 & -1 & -2 \\ -2 & -2 & -4 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 1 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad \xi_1 = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} \quad \xi_2 = \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix}$$

$$(6 \cdot E - C)x = 0 \quad \begin{pmatrix} 5 & -1 & -2 \\ -1 & 5 & -2 \\ -2 & -2 & 2 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & -5 & 2 \\ 0 & 0 & 0 \\ 0 & 2 & -1 \end{pmatrix} \quad \xi_3 = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$$

将 $\xi_1 = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}$ $\xi_2 = \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix}$ $\xi_3 = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$ 正交化单位化

因为取 ξ_1, ξ_2 是正交的

正交矩阵 $Q = \begin{pmatrix} \frac{-1}{\sqrt{2}} & \frac{-1}{\sqrt{3}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{3}} & \frac{1}{\sqrt{6}} \\ 0 & \frac{1}{\sqrt{3}} & \frac{2}{\sqrt{6}} \end{pmatrix}$

正交变换 $x = Qy$, 将 $f(x_1, x_2, x_3)$ 化为 $6y_3^2$.